

深度学习实践 论文复现报告



项目中文名称: 残差去噪扩散模型

项目英文名称: Residual Denoising Diffusion Models

答辩人: 李毅 PB22051031

院 系: 信息科学技术学院

时 间: 2025 年 6 月 24 日

第一部分 文章摘要和介绍翻译

本报告基于论文: Residual Denoising Diffusion Models, 已经发表在 CVPR 2024 上。

1.1 摘要翻译

我们提出残差去噪扩散模型 (RDDM), 一种新颖的双重扩散过程, 它将传统的单一去噪扩散过程解耦为残差扩散和噪声扩散。这种双重扩散框架通过引入残差, 将最初在图像修复中不可解释的基于去噪的扩散模型, 扩展为适用于图像生成和修复的统一且可解释的模型。具体而言, 我们的残差扩散表示从目标图像到退化输入图像的定向扩散, 并显式地指导用于图像修复的反向生成过程; 而噪声扩散则代表了扩散过程中的随机扰动。残差优先考虑确定性, 而噪声强调多样性, 这使得 RDDM 能够有效统一处理具有不同确定性或多样性需求的任务, 例如图像生成和修复。我们通过系数变换证明了我们的采样过程与 DDPM 和 DDIM 的一致性, 并提出了一种部分路径无关的生成过程以更好地理解反向过程。值得注意的是, 我们的 RDDM 使得仅用 L1 损失、批大小为 1 训练的通用 UNet, 能够与最先进的图像修复方法相媲美。我们提供代码和预训练模型, 以鼓励对本创新框架的进一步探索、应用和开发。

We propose residual denoising diffusion models (RDDM), a novel dual diffusion process that decouples the traditional single denoising diffusion process into residual diffusion and noise diffusion. This dual diffusion framework expands the denoising-based diffusion models, initially uninterpretable for image restoration, into a unified and interpretable model for both image generation and restoration by introducing residuals. Specifically, our residual diffusion represents directional diffusion from the target image to the degraded input image and explicitly guides the reverse generation process for image restoration, while noise diffusion represents random perturbations in the diffusion process. The residual prioritizes certainty, while the noise emphasizes diversity, enabling RDDM to effectively unify tasks with varying certainty or diversity requirements, such as image generation and restoration. We demonstrate that our sampling process is consistent with that of DDPM and DDIM through coefficient transformation, and propose a partially path-independent generation process to better understand the reverse process. Notably, our RDDM enables a generic UNet, trained with only an L1 loss and a batch size of 1, to compete with state-of-the-art image restoration methods. We provide code and pre-trained models to encourage further exploration, application, and development of our innovative framework (<https://github.com/nachifur/RDDM>).

1.2 介绍翻译

在现实场景中, 扩散常以涉及多重并行过程的复杂形式出现, 例如多种气体的弥散或不同类型波/场的传播。这促使我们思考: 基于去噪的扩散模型是否因仅聚焦于去噪而存在局限性。当前基于扩散的图像恢复方法通过将退化图像作为条件输入隐式指导反向生成过程, 从而将扩散模型扩展至图像恢复任务, 但未改变原始的去噪扩散过程。然而, 从噪声开始的反向过程似乎是不必要的, 因为退化图像本身是已知的。正向过程对图像恢复缺乏可解释性, 如图 1a 所示, 因其扩散过程不包含任何关于退化图像的信息。

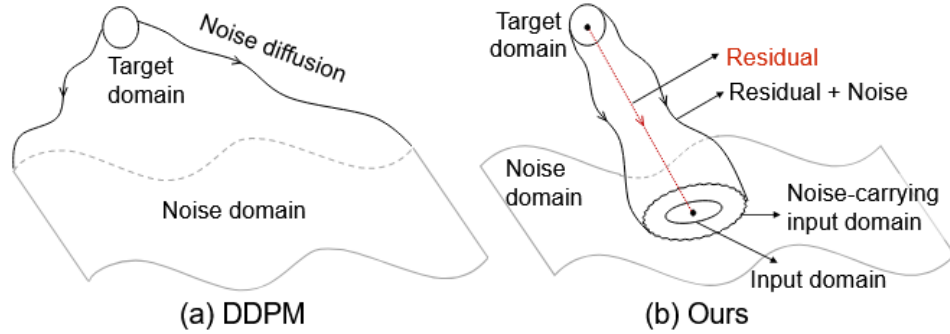


图 1: DDPM(a) 与我们的残差去噪扩散过程 (b)。针对图像恢复任务, 我们引入残差扩散以表示从目标图像到输入图像的扩散方向。

本文探索了一种新颖的双重扩散过程, 提出残差去噪扩散模型 (RDDM), 旨在解决单一去噪过程在图像恢复中的不可解释性问题。在 RDDM 中, 我们将先前的扩散过程解耦为残差扩散与噪声扩散: 残差扩散优先处理确定性, 表示从目标图像到条件输入图像的定向扩散; 噪声扩散强调多样性, 表示扩散过程中的随机扰动。因此, RDDM 能统一处理具有不同确定性或多样性需求的任务 (例如图像生成与恢复)。相较于基于去噪的扩散模型, RDDM 中的残差明确指示前向扩散方向 (如图 1b 所示), 并显式指导图像恢复的反向生成过程。

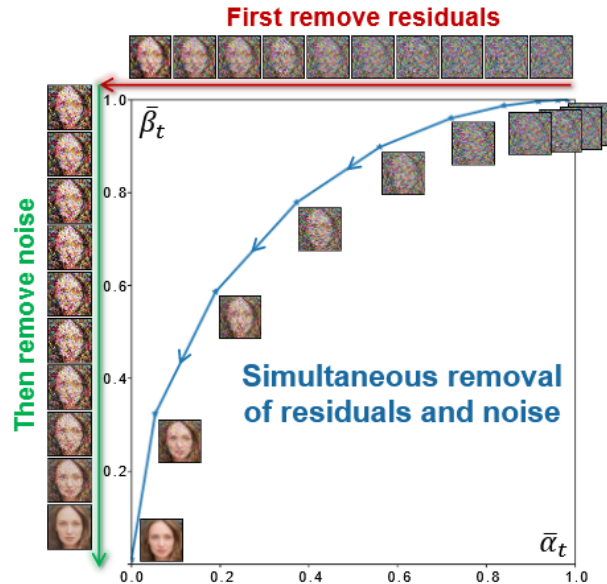


图 2: 解耦的双重扩散框架。先前的前向扩散过程被解耦为残差扩散和噪声扩散, 而在反向过程中, 同步采样可解耦为先去除残差再去除噪声。

具体而言，我们重新定义允许残差与噪声同步扩散的新前向过程——目标图像逐渐扩散为纯噪声图像（用于图像生成）或携带噪声的输入图像（用于图像恢复）。与传统去噪扩散模型使用单一系数调度方案控制噪声和图像的混合比例不同，RDDM 采用两个独立系数调度方案分别控制残差与噪声的扩散速度。我们发现这种独立性在反向生成过程中同样显著：测试阶段在一定范围内调整系数调度方案不影响图像生成结果；优先消除残差再进行去噪（见图 2）也能生成语义一致的图像。RDDM 与主流去噪扩散模型兼容：通过系数变换，其采样过程与 DDPM 及 DDIM 一致。此外，RDDM 原生支持条件输入，使仅用 L1 损失函数和批大小为 1 训练的模型能够与最先进的图像恢复方法竞争。我们期望该模型能推动建立统一可解释的图像到图像分布变换方法论，并强调残差与噪声在扩散模型中的同等重要性——残差主导确定性，噪声主导多样性。本文贡献如下：

- (1) 提出新型双重扩散框架，通过引入残差解决单一去噪过程在图像恢复中的不可解释性问题。我们的残差扩散表示从目标图像到条件输入图像的定向扩散。
- (2) 引入部分路径无关的生成过程，解耦残差与噪声，分别突出其在控制定向残差偏移（确定性）与随机扰动（多样性）中的作用。
- (3) 设计自动目标选择算法，为未知新任务选择预测残差或噪声。
- (4) 大量实验表明，该方法可适配不同任务（如图像生成、恢复、修复及转换），满足确定性或多样性需求，并支持配对与非配对数据。

In real-life scenarios, diffusion often occurs in complex forms involving multiple, concurrent processes, such as the dispersion of multiple gases or the propagation of different types of waves or fields. This leads us to ponder whether the denoising-based diffusion models have limitations in focusing solely on denoising. Current diffusion based image restoration methods extend the diffusion model to image restoration tasks by using degraded images as a condition input to implicitly guide the reverse generation process, without modifying the original denoising diffusion process. However, the reverse process starting from noise seems to be unnecessary, as the degraded image is already known. The forward process is non-interpretability for image restoration, as the diffusion process does not contain any information about the degraded image, as shown in Fig. 1a.

In this paper, we explore a novel dual diffusion process and propose Residual Denoising Diffusion Models (RDDM), which can tackle the non-interpretability of a single denoising process for image restoration. In RDDM, we decouple the previous diffusion process into residual diffusion and noise diffusion. Residual diffusion prioritizes certainty and represents a directional diffusion from the target image to the conditional input image, and noise diffusion emphasizes diversity and represents random perturbations in the diffusion process. Thus, our RDDM can unify different tasks that require different certainty or diversity, e.g., image generation and restoration. Compared to denoisingbased diffusion models for image restoration, the residuals in RDDM clearly indicate the forward diffusion direction and explicitly guide the reverse generation process for image restoration, as shown in Fig. 1b.

Specifically, we redefine a new forward process that allows simultaneous diffusion of residuals and noise, wherein the target image progressively diffuses into a purely noisy image for image generation or a noise-carrying input image for image restoration. Unlike the previous denoising diffusion model, which uses one coefficient schedule to control the mixing ratio of noise and images, our RDDM employs two independent coefficient schedules to control the diffusion speed of residuals and noise. We found that this independent diffusion property is also evident in the reverse generation process, e.g., readjusting the coefficient schedule within a certain range during testing does not affect the image generation results, and removing the residuals firstly, followed by denoising see Fig. 2, can also produce semantically consistent images. Our RDDM is compatible with widely used denoising diffusion models, i.e., our sampling process is consistent with that of DDPM and DDIM by transforming coefficient schedules. In addition, our RDDM natively supports conditional inputs, enabling networks trained with only an ℓ_1 loss and a batch size of 1 to compete with state-of-the-art image restoration methods. We envision that our models can facilitate a unified and interpretable image-to-image distribution transformation methodology, highlighting that residuals and noise are equally important for diffusion models, e.g., the residual prioritizes certainty while the noise emphasizes diversity. The contributions of this paper are summarized as follows:

1. We propose a novel dual diffusion framework to tackle the non-interpretability of a single denoising process for image restoration by introducing residuals. Our residual diffusion represents a directional diffusion from the target image to the conditional input image.
2. We introduce a partially path-independent generation process that decouples residuals and noise, highlighting their roles in controlling directional residual shift certainty and random perturbation diversity, respectively.
3. We design an automatic objective selection algorithm to choose whether to predict residuals or noise for unknown new tasks.
4. Extensive experiments demonstrate that our method can be adapted to different tasks, e.g., image generation, restoration, inpainting and translation, focusing certainty or diversity, and involving paired or unpaired data.